IS1203
Foundations of Information Systems

Dr. Thushani Weerasinghe
Senior Lecturer
University of Colombo School of Computing

Topic 5: Social and Ethical Issues Related to Information Systems

1. Computer-Related Wastes and Mistakes
2. Computer Crimes and Security Issues
3. Ethics in Information Society
4. Moral Dimensions of the Information Age
5. Ergonomics

Intended Learning Outcomes

- Identify possible wastes in an IS environment, their causes, and possible solutions.
- Identify policies and procedures useful in eliminating waste and mistakes.
- Explain the types and effects of computer crime.
- Identify specific measures to prevent computer crime.
- Identify five moral dimensions of
- List the important effects of computers on the work environment.
- Identify specific actions that must be taken to ensure the health and safety of employees.
- Prepare a criterion for the ethical use of information systems.

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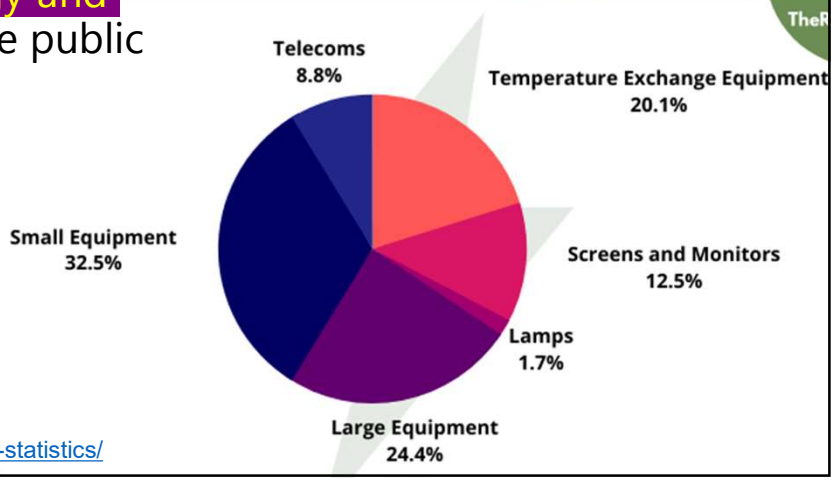
e-Waste

Inappropriate use of computer technology and resources in both the public and private sectors



https://theroundup.org/global-e-waste-statistics/

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Computer Waste

- Discard technology
- Unused systems
- **Personal use of corporate** time and IT infrastructure

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What Are The Toxic Components?

Read
https://www.sciencedirect.com/science/article/abs/pii/S0957582018300533

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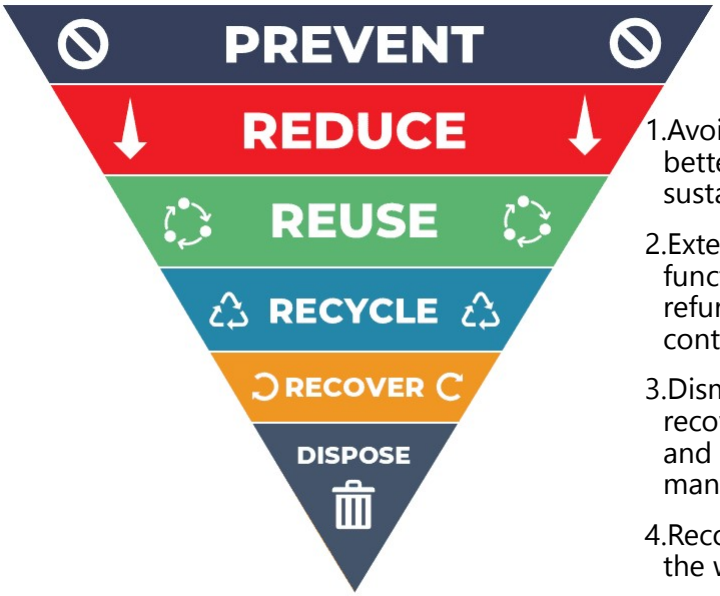

Waste Management Hierarchy

1. Avoid generating e-waste through better product design and more sustainable consumption habits
2. Extend the life of electronics and their functional components by repairing, refurbishing, or redeploying devices for continued use
3. Dismantle electronic equipment to recover valuable materials like metals and plastics, which can then be used to manufacture new products
4. Recover energy or other materials from the waste.
5. Dispose of the waste safely

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Preventing Computer Waste and Mistakes

1. **Establish Policies and Procedures for efficient acquisition, use, and disposal** of systems and devices
2. **Implement** Policies and Procedures
   - Training (for acceptance and implementation)
3. **Monitor** Policies and Procedures
   - Internal audits
4. **Review** Policies and Procedures
   - Revise current policies, & add new policies

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Computer crimes

Computer Crimes … contd.

- Identity theft
  - A crime in which an imposter obtains key pieces of personal identification information in order to impersonate someone else.
- Cyberterrorism
  - Force governments or organizations to advance his/her own political objectives by launching computer-based attacks against computers, networks, and the information stored on them.

Computer Crimes … contd.

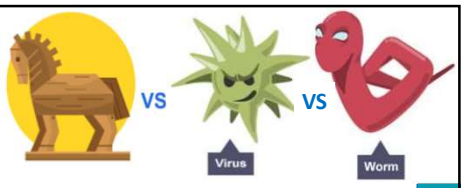
Illegal access and use

- Hacker –computer experts who studies systems to discover holes within systems and the reasons for such holes
- Cracker – attempts to break the systems by gaining unauthorized or illegal access to computer systems
- Malicious insiders – People within the organization, or business associates, who have inside information concerning security practices, data and *computer* systems and attempt to harm the systems or data

Computer Crimes … contd.

- **Information and equipment theft** –
  - Steal software and hardware
  - Steal identification numbers, passwords, invasion of privacy, plant viruses,…
- **Software and Internet piracy**
  - Act of illegally duplicating software
  - Illegally downloading software from the internet
- **Computer-related scams** – for wrongful financial or personal gains
- **Data alteration and deletion**

Data Alteration and Destruction



- **Virus**
  - Malicious programs that attach themselves to files or programs and spread when the infected files are executed.
  - Use antivirus programs such as Symantec's Norton Antivirus, and McAfee and Windows Defender and scan for virus.
- **Worm**
  - Independent program that replicates its own program files until it interrupts the operation of networks and computer systems. **Human action is not required *to* spread.**
- **Trojan**
  - Malicious software that disguises itself as a legitimate program to trick users into installing it. It has characteristics of both virus and worm.

Other Common Issues

- Ransomware –Malware that encrypts the victim's files and demands a ransom payment for the decryption key.
- Phishing - A social engineering attack where attackers impersonate legitimate entities to deceive individuals into providing sensitive information, such as login credentials or financial details. Phishing can be executed via email, SMS, or fake websites.
  - Vishing and Smishing – Voice over Phishing or SMS over Phishing
  - Spear Phishing - A targeted form of phishing where attackers customize their messages to a specific individual or organization, often using information gathered from social media.
  - Loan scams, Work from home scams, Donation or Lottery winning scams – Dial #132# from your phone to access & verify your registration details such as name, NIC number, and Mobile number. Next access more details.
  - OTP scams, be careful when purchase things via social media and making payments online.
  - Go to haveibeenpwned.com and type your email address

Other Common Issues…contd.

- Spyware- Secretly monitors and collects information via tracking keystrokes, capturing screenshots, or harvesting personal information.
- Adware- Programs that automatically deliver advertisements, as pop-ups or banners. Can be intrusive, degrade system performance and collect user data without permission.
- Rootkits- A type of malware designed to hide the presence of other malicious software on a system by modifying the operating system or other software.
- Malvertising- The distribution of malware through online advertisements. Users may be exposed to malicious code by simply visiting a website that displays compromised ads.

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Other Common Issues…Contd.

- Dumpster diving- Shred information before throwing it away. Get your details from your thrash. Never throw away your contact numbers, addresses and other personal details
- Shoulder Surfing – over the shoulder stealing credentials, never give your bank cards to another.
- Password hacking – treat password as your toothbrush
- Use a TEMP gmail account for a short duration use of an email, or else use a multi-factor authentication

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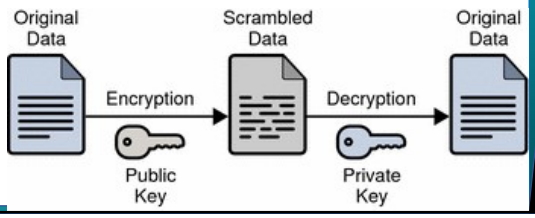
Inter-networked Security Defenses

Encryption

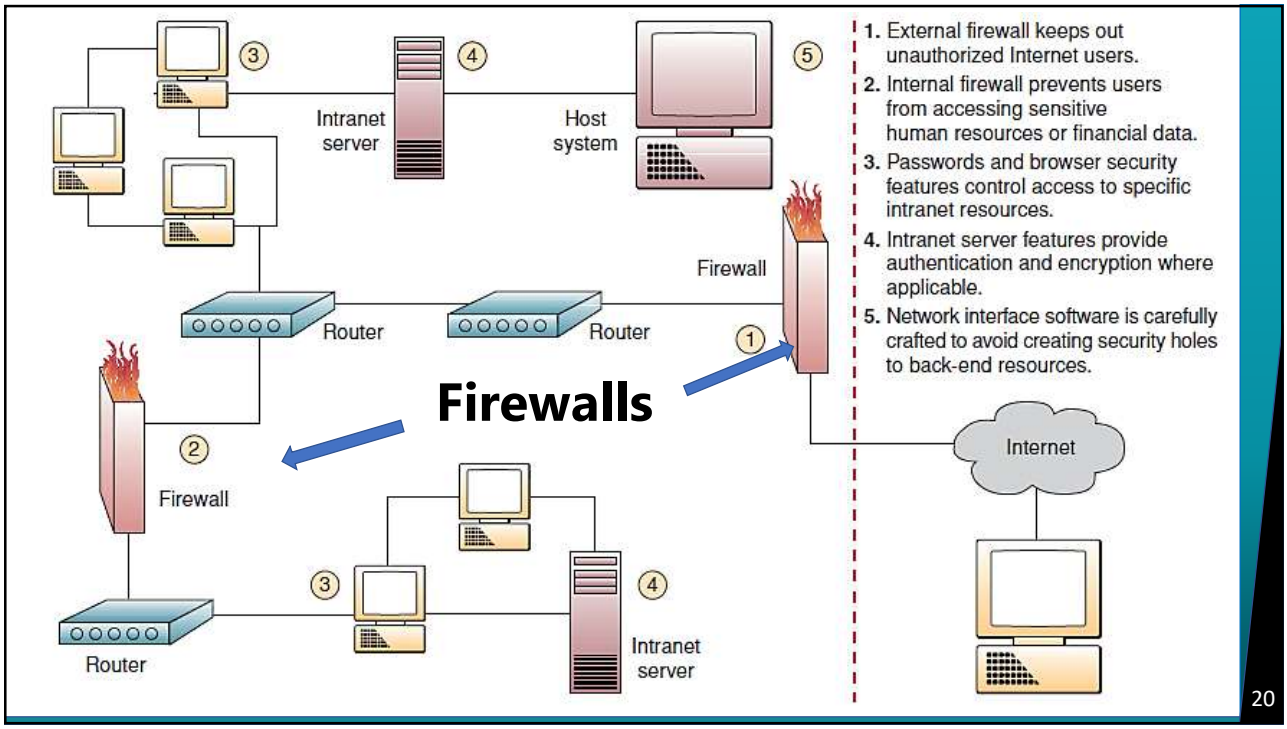
- Encryption of data has become an important way to protect data and other computer network resources, especially on the Internet, intranets, and extranets.
- Encryption involves using special mathematical algorithms, or keys, to transform digital data into an encoding before they are transmitted and decode the data when received.



Inter-networked Security Defenses…Contd.

Firewall

- A firewall serves as a gatekeeper system that protects a company's intranets and other computer networks from intrusion by providing a filter and safe transfer point for access to and from the Internet and other networks.



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Preventing Computer-Related Crime

- Biometrics – metrics related to human characteristics: physiological or behavioural
- Intrusion Detection Software – Software that monitors the system and network resources and notifies network security personnel when it senses a possible intrusion
- Managed Security Service Providers (MSSPs) - provides outsourced monitoring and management of **security** devices and systems – Contact **Sri Lanka Computer Emergency Readiness Team (SLCERT)**.
- Internet Laws –Includes legislation, regulations, and guidelines designed to address cybersecurity, data protection, and cybercrime

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Other Security Measures

- Anti-virus software
- Develop internet and security policies
- Use security codes (e.g. PIN, OPT,…etc.)
- Backup files
- Security monitors and monitor employee activities
- Biometric security
- Computer failure controls
- Fault-tolerant systems
- Disaster recovery
- Security specialists to perform audits

Activity

1. Identify six (06) potential threats that users may encounter while using social media, web browsing, and email. Describe how to protect against each of them.

2. List activities that can be done to protect corporate data.

How to Protect Your Corporate Data

1. Install strong user authentication and encryption capabilities on your firewall
2. Install the latest security patches, which are often available at the vendor's Internet site
3. Disable guest accounts and null user accounts to avoid intruders accessing the network without a password
4. Do not provide overfriendly log-in procedures for remote users
5. Restrict physical access to the server
6. Turn audit trails on to trace the detailed transactions

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How to Protect Your Corporate Data …contd.

7. Install a corporate firewall between your corporate network and the Internet
8. Install antivirus software
9. Keep all software updated on all computers
10. Conduct regular IS security audits
11. Verify and exercise frequent data backups for critical data
12. Keep yourself up to date on major security breaches.
13. Manage your social media settings.

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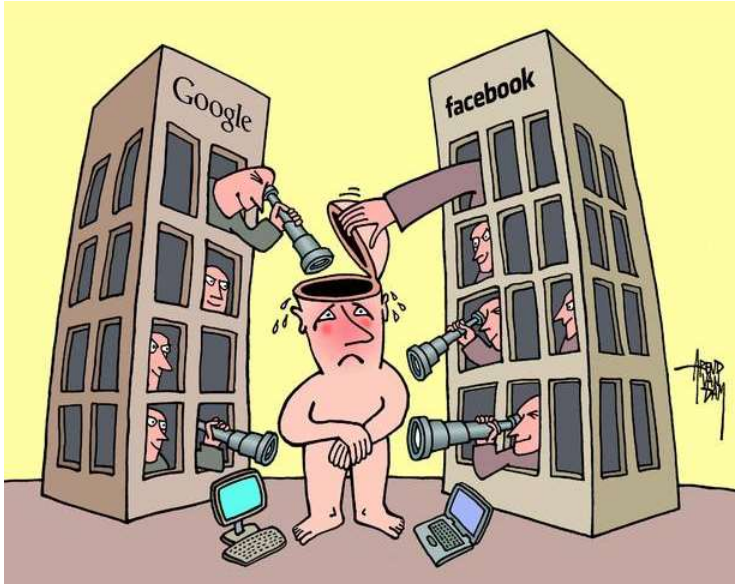
Privacy issue



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Privacy Issues

- **Privacy** is the claim of individuals to be left alone, free from surveillance or interference from other individuals or organizations
- With IS, privacy deals with the collection and use or misuse of data distributed over a network
- E.g., Government databases collecting genetic, medical and lifestyle data from middle-aged people
  - Enable doctors to discover new treatment
  - Concerns – data used for commercial exploitation (e.g. insurance companies)
- Open data concept in SL

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Privacy issues

- Privacy at work
  - Monitor every key stroke on the keyboard
  - Monitor number of breaks
- E-mail privacy
  - Monitor e-mail sent and received by employees
  - Erased e-mails can be retrieved and used in lawsuits
- Privacy and the Internet
  - Access to criminal hackers, marketing companies, corporate bosses
  - Policy statements on what information will be collected and what is done with it

Ethical Issues in Information Systems

Ethics refers to the principles of right and wrong that individuals, acting as free moral agents, use to make choices to guide their behaviours. E.g.;

- Many companies use information technology to reduce the sizes of their workforces.
- Voice recognition software reduces the need for human operators by enabling computers to recognise a customer's responses to a series of computerised questions.
- Many companies monitor what their employees are doing on the Internet to prevent them from wasting company resources

Ethics- Basic Concepts

- **Responsibility** is a key element of ethical action. Responsibility means that you accept the potential costs, duties, and obligations for the decisions you make.
- **Accountability** is a feature of systems and social institutions: It ensures that the individuals or organization act responsibly, maintain integrity, and contribute positively to society. As such, the IT and systems are in place to determine who took the respective action and who was responsible.
- **Liability** is a feature of political systems in which a body of laws is in place that permits individuals to recover the damages done to them by other actors, systems, or organizations.

Technology Trends that Raise Ethical Issues

- Computing power doubles every 18 months - More organizations depend on computer systems for critical operations.
- Data storage costs are rapidly declining - Organizations can easily maintain detailed databases on individuals.
- Data analysis advances - Companies can analyze vast quantities of data gathered on individuals to develop detailed profiles of individual behaviour.
- Networking advances and the Internet - Copying data from one location to another and accessing personal data from remote locations are much easier.

Work environment Issues

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Health Concerns

- Occupational stress: job security, loss of control, demotion, incompetence
  - Damaged computer equipment
  - Monitor stress levels and provide training and counseling
- Physical concerns: Repetitive strain injury (RSI)
  - Pain felt in muscles, nerves and tendons caused by repetitive movement and overuse

- Repetitive strain injury is one possible cause of carpal tunnel syndrome. Repeated hand and wrist movements can irritate the wrist tendons.

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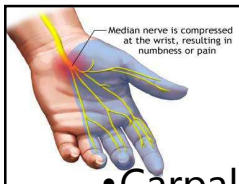
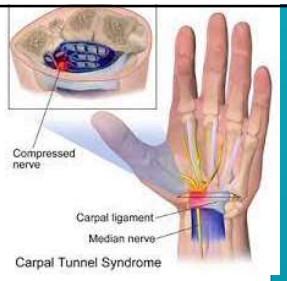
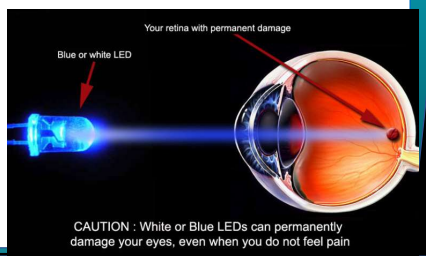


Health concerns …Contd.



- Carpal tunnel syndrome (CTS)
- Computer Vision Syndrome-
  - due to prolonged computer use. E.g., Eye irritation (Dry eyes, itchy eyes, red eyes) Blurred vision & Headaches.
  - and due to emissions from improperly maintained and used equipment
- Shoulder, neck and back pain



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ACTIVITY

- What is Techno Stress?
- How can computer science and information system students get it?
- How to overcome it?
  - Set Boundaries
  - Take Breaks
  - Practice Mindfulness
  - Limit Notifications
  - Seek Support
  - Technology Management

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Ergonomics

- The study of designing and positioning computer equipment for employee health and safety.
- This concerns;
  - Slope of keyboard
  - Positioning and designing of display screens
  - Placement of display screens
  - Placement and design of computer tables and chairs

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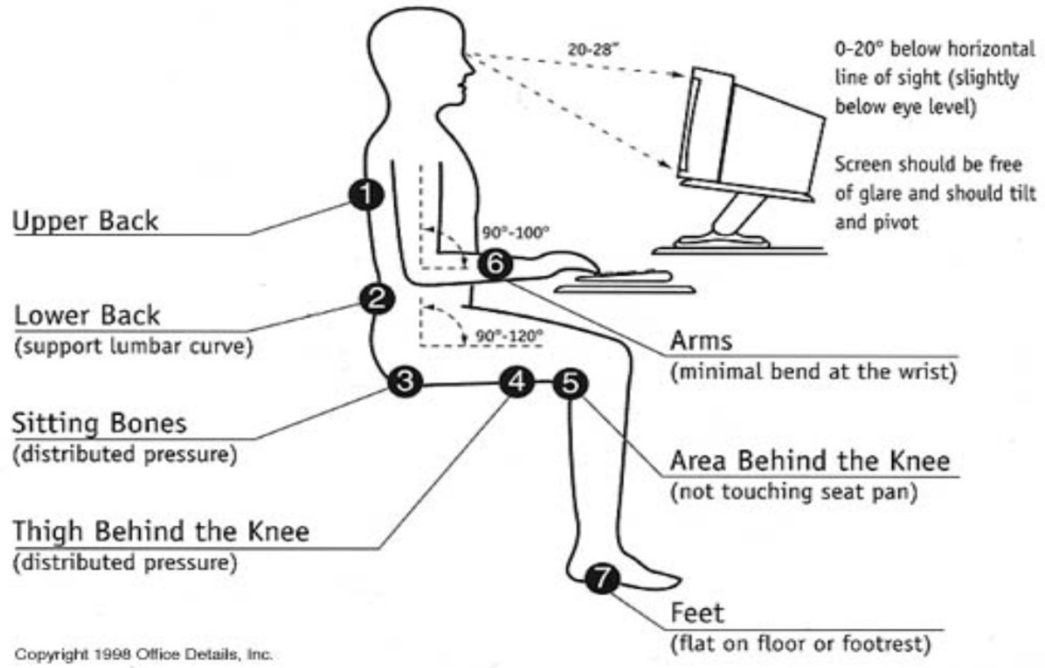

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Avoiding Health and Environment Problems

- Maintain good posture and positioning.
- Don't ignore pain or discomfort.
- Use stretching and strengthening exercises.
- Find a good physician who is familiar with injuries such as Repetitive strain injury (RSI) and how to treat it.

Exercise:

1. Identify five strong measures organizations can take to safeguard against the theft of computer hardware and the data stored on it.
2. What is a virus? What is a worm? How are they different?
3. What is ergonomics? How can it be applied to office workers?
4. What specific actions can you take to avoid spyware?